THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** mouth, oesophagus, stomach, small intestine, large intestine	Standard order.
2	**(b)** 960	16 x 60.
3	**(d)** four chambers	Mammal.
4	**(a)** starch is delayed	Amylase missing.
5	**(c)** increases	More oxygen needed.
6	**(c)** arteries	Goes to lungs.
7	**(d)** breaking fat into tiny droplets	Emulsification.
8	**(b)** decreases	Air goes out.
9	**(c)** veins	Veins.
10	**(c)** pepsin: proteins	Pepsin acts on proteins.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Muscles need more energy and oxygen (1); More carbon dioxide has to be removed (1).	1+1
12	Arteries have thick elastic walls (1); Veins have thin walls and valves (1).	1+1
13	More water is lost as sweat (1); Kidneys conserve water (1).	1+1
14	Smaller pieces digest faster (1); More saliva mixes (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mouth: starch begins digestion (1); Stomach and small intestine: digestion completed (1); Large intestine: water removed, undigested egested (1).	1+1+1
16	Damages lungs (1); Causes cough and cancer (1); Reduces stamina (1).	1+1+1
17	Constipation (1); Acidity (1); Indigestion (1).	1+1+1
	OR: Blow through a straw into lime water (1); It turns milky (1); CO2 is present (1).	1+1+1
18	Spiracles open to the outside (1); Tracheae carry air (1); Direct to cells (1).	1+1+1
	OR: Gills need water (1); Cannot take oxygen from air (1); Gills collapse (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mouth: amylase on starch (1); Stomach: pepsin on proteins (1); Small intestine: bile for fat, pancreatic juice for all (1); End products: sugar, amino acids, fatty acids (1); Absorbed by villi (1).	1+1+1+1+1
	OR: Humans: lungs (1); Fish: gills (1); Insects: tracheae (1); Earthworm: moist skin (1); Amoeba: surface of the body (1).	1+1+1+1+1
20	Four chambers (1); Right side deoxygenated (1); Lungs oxygenate (1); Left side pumps to body (1); Blood returns (1).	1+1+1+1+1
	OR: Heart stronger (1); Lungs larger capacity (1); Kidneys: enough water (1); Fibre for gut (1); Avoid junk food (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that bile contains enzymes; bile only helps in digesting **fats** and is made by the liver.
- Writing that arteries carry blood to the heart and veins from the heart; it is exactly the **opposite**.
- Confusing ingestion, digestion, absorption and egestion, especially calling egestion 'excretion'.
- Mixing up breathing (taking air in and out) with respiration (release of energy from food in cells).
- Writing that the large intestine digests food; it mainly **absorbs water**.
- In the breathing movement, saying the diaphragm moves up during inhalation; it moves **down**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** oxygen	Oxygen is used.
2	**(a)** anaemia	Less haemoglobin.
3	**(d)** large intestine	Reabsorbs water.
4	**(b)** exhaled air has more carbon dioxide	Test for CO2.
5	**(a)** white blood cells	Defence.
6	**(d)** fats	Fat digestion.
7	**(c)** carbon dioxide, water and energy	Aerobic.
8	**(b)** 72	Normal pulse.
9	**(c)** liver	Liver is a gland.
10	**(d)** alveoli	Thin walls.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Less oxygen reaches cells (1); Less energy released (1).	1+1
12	Lungs: carbon dioxide (1); Kidneys: urea (1).	1+1
13	They have a large surface area (1); They are rich in blood vessels (1).	1+1
14	It breaks fat into small droplets (1); Enzymes can then digest the fat easily (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Absorbed in intestine and lungs (1); Blood carries them (1); Capillaries supply cells (1).	1+1+1
16	Separate chambers (1); Walls divide (1); Efficient supply (1).	1+1+1
17	Stomach churns and digests proteins (1); Intestine completes digestion (1); Intestine absorbs (1).	1+1+1
	OR: They kill germs (1); More are made (1); Defence system active (1).	1+1+1
18	Heart beats faster (1); Pumps more blood (1); Muscles need oxygen (1).	1+1+1
	OR: Clot blood (1); Stop bleeding (1); Prevent infection (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Inhalation: diaphragm down, ribs up (1); Volume increases (1); Pressure falls (1); Air enters (1); Exhalation: reverse (1).	1+1+1+1+1
	OR: Rest: about 72 (1); Walking: higher (1); Running: highest (1); Muscles need oxygen (1); Heart pumps faster (1).	1+1+1+1+1
20	Carries materials to all cells (1); Anaemia: fewer RBC (1); Less oxygen reaches cells (1); Heart failure stops circulation (1); Care and regular checkups (1).	1+1+1+1+1
	OR: Makes bile (1); Stores extra food (1); Breaks down harmful substances (1); Processes digested food (1); Is the largest gland (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** increases	More oxygen needed.
2	**(c)** increase surface area for absorption	More area.
3	**(a)** lungs	Respiratory system.
4	**(c)** capillaries	Exchange of materials.
5	**(a)** muscular contractions (peristalsis)	Waves.
6	**(c)** cockroach	Insect.
7	**(b)** capillaries	Largest artery.
8	**(d)** kill germs and activate pepsin	Helps proteins.
9	**(b)** earthworm	Moist skin.
10	**(c)** arteries carry blood away from the heart	Standard.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Oxygen dissolves in moisture (1); Gas exchange occurs (1).	1+1
12	Easy exchange of gases (1); Oxygen diffuses into blood (1).	1+1
13	Microbes in its rumen digest it (1); Chews cud (1).	1+1
14	It carries oxygen, food and wastes (1); It reaches every part of the body (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Breathing is physical (1); Respiration is chemical (1); Occurs in lungs vs in cells (1).	1+1+1
16	Secretes pancreatic juice (1); Digests starch, proteins, fats (1); Poured into small intestine (1).	1+1+1
17	Grass is partly digested (1); Chewed again (1); Cellulose digested (1).	1+1+1
	OR: Filter wastes from blood (1); Make urine (1); Water helps flush wastes (1).	1+1+1
18	Waste leaves through the body surface (1); Undigested food is thrown out through the cell surface (1); There are no special organs for excretion (1).	1+1+1
	OR: Sweat removes salts (1); Evaporation cools (1); Skin is an organ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Amoeba: food vacuole (1); Cow: rumen and cud (1); Human: alimentary canal (1); Glands in humans (1); Different foods (1).	1+1+1+1+1
	OR: Capillaries thin walled (1); Exchange with cells (1); Food and oxygen in (1); Wastes out (1); Veins carry blood back (1).	1+1+1+1+1
20	Urea from kidneys (1); CO_2 from lungs (1); Sweat from skin (1); Wastes are toxic (1); Water balance (1).	1+1+1+1+1
	OR: Fish gills have many thin filaments (1); Water flows over the gills (1); Oxygen dissolved in water enters the blood (1); Earthworm has a moist skin (1); Oxygen dissolves in the moisture of the skin (1).	1+1+1+1+1

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